DAY 1 OF 4

Streams and Conversion

Your first working patch — in one hour

By the end of today you will have a live webcam effect running.

It will be ugly. That is the point — ugly and working beats pretty and theoretical.

One idea before we touch anything

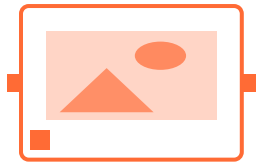
> ***TouchDesigner is streams of data, and boxes that convert between them.***

That is genuinely most of it. The rest of the week is learning which boxes exist and how to plumb them together.

- **Nothing runs top to bottom** — There is no script executing line by line. Data flows left to right through whatever you have wired up, sixty times a second, forever.
- **A box is a verb** — Every node does one thing to what comes in and passes the result on. Blur. Add. Select. Smooth.
- **The craft is conversion** — Turning a picture into numbers, numbers into movement, movement into a picture again. That is the whole job.

Four families you will actually use

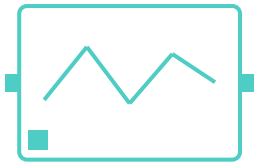
Every node belongs to one. Each family looks different in the network.



TOP stands for ***Texture Operator***

Images

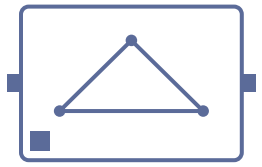
Webcam, video, blur, colour. Its viewer shows the image itself.



CHOP stands for ***Channel Operator***

Numbers over time

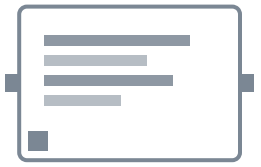
A slider, a fingertip position, audio. Its viewer shows a moving line graph. This is where interaction lives.



SOP stands for ***Surface Operator***

3D geometry

Points, lines and shapes in space. Its viewer shows a wireframe. Day 3 is built on these.



DAT stands for ***Data Operator***

Text and tables

Notes, lists, Python. Its viewer shows rows of text.

*Two more exist: **COMP** (Component — a box holding a whole network inside it) and **MAT** (Material — how a 3D surface looks). Both arrive later this week.*

Three moves and you can build anything

Learn these now. Everything else is variations.

1

Press Tab

Make a node

Click empty space, press Tab, type the node name, hit Enter. That is how every node gets made. If you cannot find a node, check the family tabs along the top of the search box.

2

Drag wire to wire

Connect them

Drag from the stub on a node's right edge to the stub on the next node's left edge. Output to input, always that direction.

3

Click the corner square

See what it does

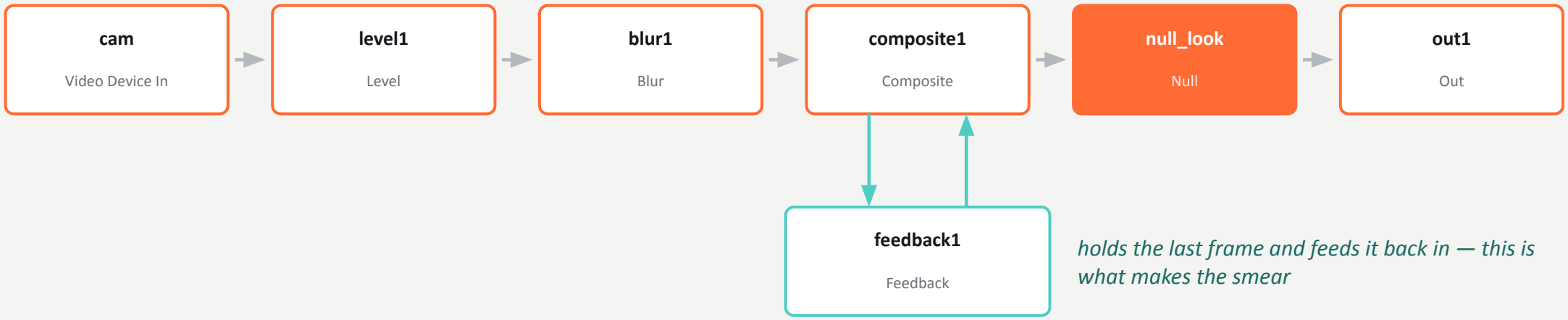
Bottom-LEFT of every node shows you its output. Without this you are working blind. (The bottom-RIGHT square is something else — we meet it on Day 3.)

Also worth knowing

Scroll to zoom, middle-drag to pan, and press H to fit everything on screen when you get lost.

What we are building

Six nodes. Each one does exactly one thing to the picture.



cam your webcam, live

level1 brightness and contrast

blur1 softens it

composite1 mixes this frame with the last one

null_look the labelled outlet — everything downstream reads this

out1 sends the result out of the container

Build it with me

Node names are what you type after pressing Tab.

1 `videodevicein`

Your camera. Click its viewer square — you should see yourself. Black means another app has the camera: quit Zoom, Teams, browser tabs, the Camera app.

2 `level`

Wire the camera into it. Push Brightness and Contrast around until it looks wrong, then back off.

3 `blur`

Wire Level into it. Set Size to about 4. Small is fine.

4 `composite`

Wire Blur into its FIRST input. Nothing interesting yet.

5 `feedback`

Wire Composite into Feedback, then Feedback back into Composite's SECOND input. Yes, a loop. Watch what happens.

6 `null`

Wire Composite into it. Rename it null_look. It changes nothing — we will talk about why it matters.

Why we added a node that does nothing

Think of a Null as a labelled outlet on the wall.

Without one

Ten things plug straight into your Blur. You add a node before it — now you rewire ten connections, and any expression naming that Blur is broken.

With one

Ten things plug into null_look. You rebuild everything behind it however you like, and nothing downstream notices. You rewired the wall; the outlet stayed put.

Where to put them

Not after every node — that is clutter. Put one at the end of each section: the point where one part hands its result to another part.

Give it a real name. null_look, null_hand, null_particles.

Address, not a wire

The second way things connect in TouchDesigner. You will meet this every single day.

Some connections are not cables. They are an address typed into a parameter field.

Wires

Data flowing from one node into the next. Left to right. This is what you built today — six nodes, five wires.

Addresses

An assignment: 'use THAT node for this setting.' You drag a node onto a parameter field, or type its path. No cable appears. Try it now: drag any node onto another node's parameter and watch the field fill in.

Why this matters

Next week your fingertip drives a parameter this way. On Week 3 a material and a camera get assigned this way. Every time it comes up I will say "address, not a wire" — and a wrong address fails SILENTLY, with no error at all, so it is always the first thing to check when nothing happens.

When everything suddenly looks like a postage stamp

Your picture collapses to a tiny blurry square and nothing you change fixes it.

This will happen to you. It is not your fault, and the symptom tells you nothing about the cause.

What happened

A Feedback node starts life at 128x128 pixels. Because it loops back into your Composite, it drags the resolution of everything downstream down to match it.

The fix

On the Composite, set Output Resolution to Custom and point its width and height at the Blur before it. The whole chain snaps back to full size.

The habit to take away: when something looks wrong, click through your chain node by node and check the resolution readout at each step. The place where it changes is the place where the problem is. This method solves more bugs this week than any specific fix.

Homework: break it three ways

Fifteen minutes. The goal is curiosity, not correctness.

1

`Blur → Size`

Push it to 100. Then to 0. Notice how the whole feel changes from one number.

2

`The feedback amount`

Take it up until the image never fades. Then find where it just barely holds a trail.

3

`Composite → Operate`

Change over to difference, then multiply, then add. Same maths as photo-editing blend modes.

Post one screenshot of your ugliest result before the end of the week

Not your prettiest. Ugly means you pushed a parameter past where it was comfortable, and that is how you find out what it actually does.

Next week

Week 2 — Motion becomes number

We plug in MediaPipe and turn your hand into a stream of numbers. By the end of the hour you will be pulling your fingertip's position out as three live values, ready to drive anything.

Before then

- [ ] Save your file. We build on it every day this week.
- [ ] Post your ugly screenshot.
- [ ] Make sure the MediaPipe example still opens and tracks your hand.
- [ ] Close Zoom and anything else using your camera before we start.

Stuck on anything from today? Reach out to me!